SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

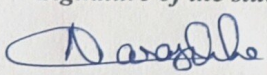
Annexure A

Descriptive Written Test

Code of Conduct: (192472344)

I R. Lakshmi Narasimha *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
   Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Intelligent Agents in a Smart Home Automation System

Introduction: An Intelligent agent perceives its environment through sensors and performs actions to acheive desired results. In a Smart home, it Controls lighting, temprature, & security based on motion, environmental Condition time & user Preference.

Types of intelligent agents

(1) Simple Reflex Agents:-

=> Acts based on the Current Input Using Condition action rule

=> Example:- If motion is detected, Switch on the lights. If no motion is detected, Switch them off

=> Advantages: Simple, fast, and easy to Implement

=> limitation: Cannot remember Past events or learn User Preferences.

(2) Model-based Agents:-

=> Maintains an Internal model of the environment & uses Past Observation to understand Current Situation

=> Example:- Remembers that a room is occupied even when the motion sensor temporarily detects no movement.

=> Advantages: Works better in Partially Observable environments

=> limitations: Requires memory & an

(3) Goal-Based Agent

⇒ Selects actions based on specific goals.

⇒ Example: Adjusts the thermostat to maintain desired room temperature & activates security when the house is unoccupied.

⇒ Advantages: Can plan actions to achieve desired outcomes

⇒ Limitation: Planning may require additional time & computation.

(4) Utility-Based Agent:

⇒ Evaluates different actions using a function & chooses the action with the highest expected benefit.

⇒ Example: Balances comfort, electricity consumption & security when deciding temperature settings & lighting schedules.

⇒ Advantages: Can balance multiple competing objectives

⇒ Limitations: Requires a suitable utility function and reliable information.

Most effective approach:

-A hybrid-model based & Utility-based agent, Suppo
by goal-based planning & simple reflex rules, would be
most effective.

=> Model-based Component: Maintains information about occup
& environmental Conditions.

=> Goal-based Component: Works toward goals such as maint
Comfortable temperatures & Securing the home.

=> Utility-based Component: Balances Comfort, Energy effici
& Security.

Conclusion: -A hybrid Intelligent agent Provides r
responses, Informed decision-making, Personalization, & a
better balance between Comfort, energy efficiency & security.

2) Solving an Unknown Maze Using UCS & IDS

Introduction:- A Robot placed in an Unknown maze find a Path from the Starting Point to the exit without knowing the Complete layout. Uninformed Search algorithms explore possible paths without using heuristic knowledge about how close a state is to the goal.

A) Uniform Cost Search (UCS):-

UCS expands in an Unknown maze must find a Path from the Starting Point to the exit without knowing the Complete layout. Uninformed Search algorithms explore possible paths without using heuristic about how close a state is to the goal.

Working steps

=> Place the Starting Position in a Porority queue
=> Remove the Position with the lowest Path Cost
=> Explore its Unvisited neighboring cells.
=> Add Valid neighbours with their Cumulative Costs
=> Repeat Until the exit is reached
=> Trace the Parent nodes to Obtain the Path.

Relationship to Intelligent agents

Agent type	Role in maze navigation
Simple reflex agent	Moves according to current sensor input
Model-based agent	Remembers explored cells, obstacles & previous positions.
Goal-based agent	Chooses actions to reach the exit
Utility based agent	Compares routes using cost, risk or other preferences.

(iii) Most-effective Combination

A model-based, goal based agent using UCS is suitable when maze movements have different costs.

→ the model-based component stores the discovered maze layout

→ the goal-based component directs the robot toward the exit.

→ UCS selects the least-cost route using accumulated movement costs

(B) Iterative Deepening Search (IDS):

IDS repeatedly performs depth-limited depth-first search, increasing the depth limit after each unsuccessful iteration.

Working Steps:

1) Start with depth limit 0
2) Search the maze up to that depth
3) If the exit is not found, increase the limit by 1
4) Repeat the search with the new limit
5) Stop when the exit is found.

c) Time & Space Complexity:

let b be the branch factor, d the depth of the shallowest solution, C^* the optimal solution cost, & ε the minimum positive step cost.

feature	UCS	IDS
Time Complexity	$O(b^{1+\lfloor C^*/\varepsilon \rfloor})$ Under standard assumption	$O(b^d)$
Space Complexity	Exponential; stores frontier	$O(bd)$
Complete?	Yes, with positive minimum steps Cost & finite branching	Yes, the final branching
Optimal?	Yes, for non-negative Costs Under Standard Conditions	Yes, when all step Costs are